

# Claudio Roncoli

ASSOCIATE PROFESSOR · KU LEUVEN

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VISITING PROFESSOR · AALTO UNIVERSITY

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🏠 <https://claudioroncoli.github.io> | 🎓 Claudio Roncoli



## Research interests

- Modelling and control of traffic systems in presence of automated and connected vehicles
- Integration, management, and optimisation of emerging mobility systems and technologies
- Optimisation of Intelligent Transportation Systems for smart and sustainable mobility
- Traffic flow theory, modelling, and simulation

## Professional positions

### Associate Professor

MOBILITY AND INDUSTRIAL MANAGEMENT (MIM), FACULTY OF ENGINEERING SCIENCE, KU LEUVEN

11/2024 – current  
Leuven, Belgium

### Visiting Professor

DEPARTMENT OF BUILT ENVIRONMENT, SCHOOL OF ENGINEERING, AALTO UNIVERSITY

10/2024 – current  
Espoo, Finland

### Associate Professor

DEPARTMENT OF BUILT ENVIRONMENT, SCHOOL OF ENGINEERING, AALTO UNIVERSITY

12/2022 – 10/2024  
Espoo, Finland

### Visiting Researcher

SCHOOL OF CIVIL ENGINEERING, UNIVERSITY OF SYDNEY

10/2023 – 12/2023  
Sydney, Australia

### Visiting Scholar

SCHOOL OF CIVIL ENGINEERING, UNIVERSITY OF QUEENSLAND

09/2023 – 10/2023  
Brisbane, Australia

### Assistant Professor

DEPARTMENT OF BUILT ENVIRONMENT, SCHOOL OF ENGINEERING, AALTO UNIVERSITY

10/2016 – 11/2022  
Espoo, Finland

### Postdoctoral Research Associate

DYNAMIC SYSTEMS & SIMULATION LABORATORY, TECHNICAL UNIVERSITY OF CRETE

05/2013 – 09/2016  
Chania, Greece

### Visiting Student

IMT SCHOOL FOR ADVANCED STUDIES

07/2012  
Lucca, Italy

### Visiting Research Assistant

CENTRE FOR TRANSPORT STUDIES, IMPERIAL COLLEGE LONDON

01 – 07/2011 and 02 – 06/2012  
London, UK

### Research Assistant/Associate

INDUSTRIAL AUTOMATION LAB / DELAB, UNIVERSITY OF GENOVA

01/2007 – 04/2013  
Genova, Italy

## Education

### Doctoral degree in System monitoring and environmental risk management

UNIVERSITY OF GENOVA

01/2010 – 04/2013  
Genova, Italy

### Master's degree in Computer Engineering - Production and Services Management

UNIVERSITY OF GENOVA

09/2006 – 07/2009  
Genova, Italy

### Bachelor's degree in Computer Engineering

UNIVERSITY OF GENOVA

09/2003 – 09/2006  
Genova, Italy

## Participation in Research Projects

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AT KU LEUVEN

**DREEMS - Optimizing Demand-REsponsive Electric Mobility Systems under uncertainty**

01/2026 - current

FUNDED BY THE RESEARCH FOUNDATION FLANDERS - FWO

Co-Principal Investigator

**Endowed Chair on Advancing Safe and Efficient Automated Mobility**

12/2025 - current

FUNDED BY ETHIAS AS A 600 YEARS OF KU LEUVEN CHAIR

Chairholder - Principal Investigator

AT AALTO UNIVERSITY

**ALCOSTO - Adaptive and Learning Control strategies for Sustainable future Traffic Operations**

09/2022 - current

FUNDED BY ACADEMY OF FINLAND

Principal Investigator

**ACUMEN - Ai-aided deCision tool for seamless mUltiModal nEtnetwork and traffic management**

06/2023 - 05/2026

FUNDED BY THE EUROPEAN COMMISSION, THROUGH THE HORIZON EUROPE PROGRAMME

Coordinator

**SMARTEDGE - Semantic Low-code Programming Tools for Edge Intelligence**

01/2023 - 03/2026

FUNDED BY THE EUROPEAN COMMISSION, THROUGH THE HORIZON EUROPE PROGRAMME

Project Partner Lead

**AIForLEssAuto - Artificial Intelligence for Urban Low-Emission Autonomous Traffic**

01/2022 - 12/2024

FUNDED BY ACADEMY OF FINLAND

Subproject Principal Investigator

**Finest Twins - Smart Cities Centre of Excellence**

12/2019 - 11/2023

FUNDED BY THE EUROPEAN COMMISSION, THROUGH THE HORIZON 2020 PROGRAMME

Smart mobility stream leader

**ULTRA - Ubiquitous Localization, communication, and sensing infrastrucTuRe for Autonomous systems**

01/2020 - 12/2022

FUNDED BY ACADEMY OF FINLAND

Subproject Principal Investigator

**Profi4 - Competitive funding to strengthen Universities' research profile**

09/2018 - 08/2022

FUNDED BY ACADEMY OF FINLAND

Smart mobility theme Co-Principal Investigator

**CMON - Connected Mobility Now**

06/2017 - 06/2022

FUNDED BY HENRY FORD FOUNDATION FINLAND

Co-Principal Investigator

**WISE-ACT - Wider Impacts and Scenario Evaluation of Autonomous and Connected Transport**

10/2019 - 03/2022

FUNDED BY THE EUROPEAN COMMISSION, THROUGH A COST ACTION

Working Group leader

**FDMM - Future Digital Mobility Management**

01/2021 - 12/2021

FUNDED BY EIT URBAN MOBILITY

Project partner Principal Investigator

**TRA2020 - The Transport Research Arena 2020 in Helsinki**

04/2018 - 10/2020

FUNDED BY THE EUROPEAN COMMISSION, THROUGH THE HORIZON 2020 PROGRAMME

Work-package leader

**MULSIMCO - Multilevel Traffic Simulation with a Cognitive Basis**

05/2017 - 08/2018

FUNDED BY ACADEMY OF FINLAND

Faculty Investigator

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|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| <b>SIMSIG - Traffic simulation for intelligent signal control</b><br>FUNDED BY RSM LTD<br>Project leader<br>AT TECHNICAL UNIVERSITY OF CRETE                                                                    | 06/2017 – 05/2018 |
| <b>TRAMAN21 - Traffic Management for the 21st Century</b><br>FUNDED BY THE EUROPEAN COMMISSION, THROUGH AN ERC ADVANCED GRANT<br>Postdoctoral Research Associate<br>AT UNIVERSITY OF GENOVA                     | 05/2013 – 02/2018 |
| <b>TIP - Transport Integrated Platform</b><br>FUNDED BY ENI R&M - SECONDARY LOGISTICS<br>Research assistant / Research associate / Software designer and developer                                              | 01/2007 – 04/2013 |
| <b>SECTRAM - Sécurité du transport de marchandises - Sicurezza nel trasporto merci (Safety for the transport of goods)</b><br>FUNDED BY THE EUROPEAN COMMISSION UNDER ALCOTRA (2007-2013)<br>Research assistant | 04/2009 – 03/2012 |
| <b>DaGoT - European Reference Framework on Dangerous Goods Transport</b><br>FUNDED BY THE EUROPEAN COMMISSION UNDER THE SIXTH FRAMEWORK PROGRAMME<br>Research assistant                                         | 01/2007 – 02/2008 |

## Teaching and Educational leadership activities

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| AT KU LEUVEN                                                                                                                                           |                   |
| <b>Co-teacher for the Intelligent Transportation Systems course</b><br>PART OF THE MASTER OF MOBILITY AND SUPPLY CHAIN ENGINEERING                     | 09/2025 - current |
| AT AALTO UNIVERSITY                                                                                                                                    |                   |
| <b>Director for the Master's programme in Spatial Planning and Transportation Engineering (SPT) course</b><br>PART OF THE SCHOOL OF ENGINEERING        | 10/2021 – 09/2024 |
| <b>Deputy Director for the Master's programme in Spatial Planning and Transportation Engineering (SPT) course</b><br>PART OF THE SCHOOL OF ENGINEERING | 08/2019 – 09/2021 |
| <b>Responsible teacher and main instructor for the Transport Modelling course</b><br>PART OF THE SPT MASTER'S PROGRAMME                                | 09/2018 – 02/2024 |
| <b>Responsible teacher and main instructor for the Traffic Flow Theory course</b><br>PART OF THE SPT MASTER'S PROGRAMME                                | 10/2016 – 04/2024 |
| <b>Deputy responsible teacher and co-instructor for the Traffic Management course</b><br>PART OF THE SPT MASTER'S PROGRAMME                            | 10/2016 – 06/2024 |

## Participation in Educational Projects

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|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| <b>ENACT - Education network for automated and connected transport for sustainable cities</b><br>FUNDED BY FINNISH MINISTRY OF EDUCATION AND CULTURE<br>Project Partner Principal Investigator | 01/2021 – 07/2023 |
| <b>Boost I&amp;E 2.0 - Boosting Innovation and Entrepreneurship Challenge Project 2021</b><br>FUNDED BY EIT URBAN MOBILITY KIC<br>Activity Lead                                                | 01/2021 – 12/2021 |

## Editorial and organisational activities

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### COMMITTEE MEMBERSHIPS

- Member of the Standing Committee on Vehicle-Highway Automation (ACP30) for the Transportation Research Board (TRB) (2022 - 2025)
- Member of the Technical Committee for the IFAC 7.4: Transportation systems (2017 - current)
- Member of the Technical Committee for the IEEE Emerging Transportation Technology Testing (ET3) (2023 - current)
- Representative of the university members in the Governing Board for the European Partnership on Connected, Cooperative & Automated Mobility - CCAM Association (2024 - current)
- Partner Representative of the General Assembly for the European Partnership on Connected, Cooperative & Automated Mobility - CCAM Association (2021 - 2024)
- Vice-chair of the Programme Committee for the Transport Research Arena (TRA2020)
- Member of the Scientific Committee for the Symposia on Management of Future motorway and urban Traffic Systems (MFTS 2016 - 2024)
- Member of the Technical Program Committee for the IFAC Symposia on Control in Transportation Systems (CTS 2018 - 2024)
- Member of the Technical Program Committee for the IEEE International Conference on Intelligent Transportation Systems (ITSC 2017, 2020 - 2023)
- Member of the Scientific Committee and Permanent Representative for Aalto University for the European Association for Research in Transportation (hEART 2015 - 2024)
- Member of the Scientific Committee for the IEEE International Conference on Models and Technologies for Intelligent Transportation Systems (MT-ITS 2021 - 2023)
- Member of the Scientific Committee for the EURO Working Group on Transportation Meetings (EWGT 2016, 2020)
- Member of the International Program Committee for the IEEE Intelligent Vehicles Symposia (IV 2018 - 2022)
- Member of the International Program Committee for the International Conferences on Vehicle Technology and Intelligent Transport Systems (VEHITS 2015 - 2019)
- Member of the Technical Program Committee for the International Conferences on Advances in Vehicular Systems, Technologies and Applications (VEHICULAR 2017 - 2019)
- Member of the Technical Program Committee for the IEEE International Conference on System of Systems (SoSE 2012)

### EDITOR FOR INTERNATIONAL JOURNALS AND CONFERENCES

- Editorial Advisory Board Member for Transportation Research Part C: Emerging Technologies
- Associate Editor for IEEE Transactions on Intelligent Transportation Systems
- Special Issues Editor (Deputy Editor-in-Chief) for IET Intelligent Transport Systems
- Associate Editor for IET Intelligent Transport Systems
- Handling Editor for Transportation Research Record: Journal of the Transportation Research Board (TRR)
- Assistant Editor for Journal of Intelligent Transportation Systems (JITS)
- Editor for Proceedings of the Institution of Civil Engineers - Transport
- Review Process Coordinator for TRB Committee on Vehicle-Highway Automation (ACP30)
- Review Editor for Frontiers in Future Transportation
- Principal organiser and Guest Editor of the Special Issue on “Management of Future Motorway and Urban Traffic Systems” within Transportation Research Part C: Emerging Technologies
- Guest Editor of the Special Issue on “Modelling, Learning, and Control of Conventional, Cooperative and Automated Motorway and Urban Traffic Systems” within Transportation Research Part C: Emerging Technologies
- Guest Editor of the Special issue on “Sustainable Development of Intelligent Transportation System and Equipment” within Sustainable Horizons
- Guest Editor of the Special Issue on “Management of Future Motorway and Urban Traffic Systems” within IEEE Transactions on Intelligent Transportation Systems
- Guest Editor of the Special Issue on “Traffic Modelling and Control for Smart, Safe and Green Mobility” within IEEE

Open Journal of Intelligent Transportation Systems

- Guest Editor of the Special Issue on “Partially Connected and Automated Traffic Operations in Road Transportation” within Journal of Advanced Transportation
- Guest Editor of the Special issue on “Rethinking transport – towards clean and inclusive mobility” within European Transport Research Review
- Guest Editor of the Special Issue on “Communication, Positioning, and Sensing Solutions for Autonomous Vehicles” within Sensors
- Associate Editor for the IEEE International Conferences on Intelligent Transportation Systems (ITSC)
- Associate Editor for the IFAC Symposia on Control in Transportation Systems (CTS)
- Associate Editor for the IEEE Intelligent Vehicles Symposia (IV)
- Associate Editor for the Forums on Integrated and Sustainable Transportation Systems (Forum ISTS)

#### **EVENT ORGANISATION**

- Principal organiser of the 12th Symposium of the European Association for Research in Transportation (hEART 2024)
- Principal organiser of the 1st Symposium on Management of Future motorway and urban Traffic Systems (MFTS 2016)
- Technical programme co-chair of the 8th IEEE International Conference on Models and Technologies for Intelligent Transportation Systems (MT-ITS 2023)
- Member of the Organising Committee for the Transport Research Arena 2020 (TRA2020)
- Co-organiser of the invited session on “Novel methods for modeling and control of mobility and traffic systems” at the 23rd European Control Conference (ECC 2025)
- Organiser of the Breakout session on “Interactive Traffic Management for Highly Automated Vehicles: A Multidisciplinary Challenge” at the 2023 TRB Annual Automated Road Transportation Symposium (ARTS23)
- Co-organiser of the special session titled “Towards sustainable traffic via trajectory optimisation and control strategies for Connected and Automated vehicles” at the 25th IEEE International Conferences on Intelligent Transportation Systems (ITSC 2022)
- Organiser of the Breakout session on “Interactive Traffic Management for Highly Automated Vehicles” at the 2022 TRB Annual Automated Road Transportation Symposium (ARTS22)
- Organiser and moderator of the invited session on “Intelligent Transport Systems” (Nordic Perspective Summit 2020)
- Organiser and Editor of the special Track on “Freeway Traffic Control” at the 15th IFAC Symposium on Control in Transportation Systems (CTS 2018)
- Co-organiser of the open invited track on “Traffic modeling, estimation and control” at the 20th World Congress of the International Federation of Automatic Control (IFAC 2017)
- Co-organiser of the special session on “Recent advances in motorway traffic control” at the 19th IEEE International Conferences on Intelligent Transportation Systems (ITSC 2016)
- Co-organiser of the invited session on “Management and control of motorway traffic systems” at the 14th IFAC Symposium on Control in Transportation Systems (CTS 2016)
- Co-organiser of the special session on “Advances in motorway traffic control” at the 18th IEEE International Conferences on Intelligent Transportation Systems (ITSC 2015)
- Co-organiser of the special session on “Design, modelling, and control of autonomous and cooperative vehicle systems” at the 18th Euro Working Group on Transportation Meeting (EWGT 2015)
- Co-organiser of the invited session on “Optimal control of motorways” at the 20th Conference of the International Federation of Operational Research Societies (IFORS 2014)

#### **REVIEWER FOR INTERNATIONAL JOURNALS**

- Transportation Research Part C: Emerging Technologies
- IEEE Transactions on Intelligent Transportation Systems
- Transportation Research Part B: Methodological
- Transportation Research Part A: Policy and Practice
- Transportation Science
- Transportation Research Record
- IET Intelligent Transport Systems
- Automatica

- IEEE Transactions on Control Systems Technology
- Control Engineering Practice
- Annual Reviews in Control
- IEEE Intelligent Transportation Systems Magazine
- IEEE Control Systems Magazine
- International Journal of Robust and Nonlinear Control
- IEEE Systems Journal
- IEEE Control Systems Letters
- Accident Analysis and Prevention
- Journal of Advanced Transportation
- Transportmetrica A: Transport Science
- Transportmetrica B: Transport Dynamics
- Transport Policy
- Transportation Letters: The International Journal of Transportation Research
- Journal of Traffic and Transportation Engineering
- EURO Journal on Transportation and Logistics
- Mathematical and Computer Modelling of Dynamical Systems
- Journal of Intelligent Transportation Systems: Technology, Planning, and Operations
- PROMET – Traffic & Transportation
- European Journal of Transport and Infrastructure Research
- International Journal of Sustainable Transportation
- Physica A
- Advances in Mechanical Engineering

#### REVIEWER FOR INTERNATIONAL CONFERENCES

- IEEE International Conferences on Intelligent Transportation Systems (ITSC)
- IEEE Intelligent Vehicles Symposia (IV)
- IEEE Conferences on Decision and Control (CDC)
- European Control Conferences (ECC)
- IFAC World Congresses (WS)
- IFAC Symposia on Control in Transportation Systems (CTS)
- Transportation Research Board (TRB) Annual Meetings
- Symposia of the European Association for Research in Transportation (hEART)
- EURO Working Group on Transportation Meetings (EWGT)
- International Symposia on Transportation and Traffic Theory (ISTTT)
- international Conferences on Traffic and Granular Flow (TGF)
- International Conferences on Vehicle Technology and Intelligent Transport Systems (VEHITS)
- European Conference on Information Systems (ECIS)
- IEEE International Conferences on System of Systems (SoSE)
- International Conferences on System Theory, Control and Computing (ICSTCC)

## Supervision activities

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AT KU LEUVEN

**Mr. Yangxiang Yang, PhD student**

KU LEUVEN, BELGIUM - ALSO AT AALTO UNIVERSITY, FINLAND

*10/2025 – current*

**Ms. Xinhui Shan, PhD student**

KU LEUVEN, BELGIUM

*05/2025 – current*

**Mr. Matteo Paiano, PhD student**

KU LEUVEN, BELGIUM

*03/2025 – current*

**Mr. Hongwei Li, PhD student**

KU LEUVEN, BELGIUM - ALSO AT TEL AVIV UNIVERSITY, ISRAEL

*09/2025 – current*

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| <b>Mr. Hao Lyu, PhD student</b><br>SOUTHEAST UNIVERSITY, CHINA                                                                     | 10/2025 – current |
| <b>Mr. Franco Filippi, PhD student</b><br>UNIVERSITY OF SALERNO, ITALY                                                             | 01/2026 – current |
| <b>Ms. Yanqing Yang, PhD student</b><br>TONGJI UNIVERSITY, CHINA                                                                   | 05/2025 – 04/2026 |
| <b>Mr. Seshadri Naik Moode, PhD student</b><br>UNIVERSITAT POLITÈCNICA DE CATALUNYA, SPAIN                                         | 09/2025 – 01/2026 |
| AT AALTO UNIVERSITY                                                                                                                |                   |
| <b>Ms. Liva Westerback, PhD student</b><br>AALTO UNIVERSITY, FINLAND                                                               | 11/2023 – current |
| <b>Ms. Heqi Wang, PhD student</b><br>AALTO UNIVERSITY, FINLAND                                                                     | 03/2023 – current |
| <b>Mr. Muhammad Haris, PhD student</b><br>AALTO UNIVERSITY, FINLAND                                                                | 09/2022 – current |
| <b>Mr. Francesco Vitale, PhD student &amp; Postdoc</b><br>AALTO UNIVERSITY, FINLAND                                                | 01/2020 – current |
| <b>Ms. Elina Aittoniemi, Part-time PhD student</b><br>AALTO UNIVERSITY, FINLAND - ALSO AT VTT TECHNICAL RESEARCH CENTRE OF FINLAND | 11/2018 – current |
| <b>Mr. Yuanxiang Yang, PhD student</b><br>AALTO UNIVERSITY, FINLAND - MOVED TO KU LEUVEN                                           | 10/2022 – 10/2025 |
| <b>Mr. Serio Agriesti, PhD student</b><br>AALTO UNIVERSITY, FINLAND                                                                | 03/2020 – 10/2024 |
| <b>Mr. Ze Zhou, PhD student</b><br>AALTO UNIVERSITY, FINLAND                                                                       | 04/2019 – 10/2024 |
| <b>Mr. Roozbeh Mohammadi, PhD student &amp; Postdoc</b><br>AALTO UNIVERSITY, FINLAND                                               | 02/2018 – 08/2023 |
| <b>Mr. Farzam Tajdari, Part-time PhD student</b><br>AALTO UNIVERSITY, FINLAND                                                      | 10/2017 – 01/2023 |
| <b>Dr. Iisakki Kosonen, Senior researcher</b><br>AALTO UNIVERSITY, FINLAND                                                         | 04/2019 – 10/2024 |
| <b>Dr. Murat Bayrak, Postdoctoral researcher</b><br>AALTO UNIVERSITY, FINLAND                                                      | 08/2023 – 12/2025 |
| <b>Dr. Shaghayegh Vosough, Postdoctoral researcher</b><br>AALTO UNIVERSITY, FINLAND                                                | 03/2021 – 07/2025 |
| <b>Dr. Charalampos Sipetas, Postdoctoral researcher</b><br>AALTO UNIVERSITY, FINLAND                                               | 01/2021 – 01/2025 |
| <b>Dr. Ramin Niroumand, Postdoctoral researcher</b><br>AALTO UNIVERSITY, FINLAND                                                   | 08/2023 – 08/2024 |
| <b>Dr. Weiming Zhao, Postdoctoral researcher</b><br>AALTO UNIVERSITY, FINLAND                                                      | 11/2019 – 09/2022 |
| <b>Dr. Cafer Avci, Postdoctoral researcher</b><br>AALTO UNIVERSITY, FINLAND                                                        | 06/2019 – 06/2022 |

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| <b>Dr. Yu Han, Postdoctoral researcher</b><br>AALTO UNIVERSITY, FINLAND                                                                                 | 09/2017 – 10/2017 |
| <b>Mr. Jingfeng Ma, PhD student</b><br>SOUTHEAST UNIVERSITY, CHINA                                                                                      | 12/2022 – 12/2023 |
| <b>Ms. Haizhen Li, PhD student</b><br>CHANG'AN UNIVERSITY, CHINA                                                                                        | 03/2021 – 03/2022 |
| <b>Mr. Mohammad Karimi, PhD student</b><br>CONCORDIA UNIVERSITY, CANADA                                                                                 | 08/2017 – 06/2020 |
| <b>Ms. Oxana Ivanova, MSc student</b><br>AALTO UNIVERSITY, FINLAND                                                                                      | 11/2024 – 12/2025 |
| <b>Ms. Julia Latvala, MSc student</b><br>AALTO UNIVERSITY, FINLAND                                                                                      | 11/2024 – 05/2025 |
| <b>Mr. Chris Yung, MSc student</b><br>AALTO UNIVERSITY, FINLAND                                                                                         | 06/2024 – 11/2024 |
| <b>Mr. Héctor Del Pino Verona, MSc student</b><br>AALTO UNIVERSITY, FINLAND - ALSO AT UNIVERSITAT POLITÈCNICA DE VALÈNCIA, SPAIN                        | 09/2019 – 01/2020 |
| <b>Mr. Germán R. Pardo González, Summer intern</b><br>AALTO UNIVERSITY, FINLAND - ALSO AT LONDON SCHOOL OF ECONOMICS, UK                                | 06/2024 – 08/2024 |
| <b>Ms. Olena Vol, Summer intern</b><br>AALTO UNIVERSITY, FINLAND - ALSO AT UNIVERSITÉ PARIS 1 PANTHÉON-SORBONNE, FRANCE                                 | 06/2022 – 08/2022 |
| <b>Mr. Petr Anashin, Summer intern</b><br>AALTO UNIVERSITY, FINLAND - ALSO AT UNIVERSITY OF TARTU, ESTONIA                                              | 06/2022 – 08/2022 |
| <b>Mr. Petr Hajduk, MSc student</b><br>AALTO UNIVERSITY, FINLAND - ALSO AT VTT TECHNICAL RESEARCH CENTRE OF FINLAND<br>AT TECHNICAL UNIVERSITY OF CRETE | 01/2018 – 12/2018 |
| <b>Ms. Maria Kontorinaki, PhD student</b><br>TECHNICAL UNIVERSITY OF CRETE, GREECE - WITH PROF. MARKOS PAPAGEORGIOU                                     | 09/2017 – 09/2017 |
| <b>Ms. Georgia R. Iordanidou, PhD student</b><br>TECHNICAL UNIVERSITY OF CRETE, GREECE - WITH PROF. IOANNIS PAPAMICHAIL                                 | 06/2013 – 03/2017 |
| <b>Ms. Sofia Papadopoulou, MSc student</b><br>TECHNICAL UNIVERSITY OF CRETE, GREECE - WITH PROF. MARKOS PAPAGEORGIOU                                    | 01/2016 – 03/2017 |
| <b>Ms. Georgia Perraki, MSc student</b><br>TECHNICAL UNIVERSITY OF CRETE, GREECE - WITH PROF. IOANNIS PAPAMICHAIL                                       | 03/2015 – 11/2016 |
| <b>Mr. Markos Fountoulakis, MSc student</b><br>TECHNICAL UNIVERSITY OF CRETE, GREECE - WITH PROF. MARKOS PAPAGEORGIOU                                   | 03/2015 – 07/2016 |
| <b>Ms. Cecilia Pasquale, PhD student</b><br>UNIVERSITY OF GENOVA, ITALY - WITH PROF. SIMONA SACONE<br>AT UNIVERSITY OF GENOVA                           | 05/2014 – 07/2014 |
| <b>Mr. Haitam Laarabi, PhD student</b><br>UNIVERSITY OF GENOVA, ITALY - WITH PROF. ROBERTO SACILE                                                       | 07/2012 – 04/2013 |

## Invited presentations \_\_\_\_\_

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| <b>Network-wide motorway traffic control in presence of Vehicle Automation and Communication Systems</b><br>UNIVERSITY OF CALIFORNIA LOS ANGELES, USA<br>IPAM Workshop on Traffic Control, Part of the long program “New directions in mathematical approaches for traffic flow management” | 26/10/2015 |
| <b>Traffic management in the era of Vehicle Automation and Communications Systems and Smart Infrastructure</b><br>UNIVERSITY OF GLASGOW, UK<br>Invited lecture                                                                                                                              | 09/12/2015 |
| <b>Multilane traffic control in the presence of automated and connected vehicles</b><br>UNIVERSITY OF LUXEMBOURG, LUXEMBOURG<br>Workshop on Smart Mobility                                                                                                                                  | 01/06/2017 |
| <b>Integrated control of motorway bottlenecks via flow metering and lane assignment</b><br>TECHNICAL UNIVERSITY OF CRETE, GREECE<br>Invited seminar                                                                                                                                         | 12/09/2018 |
| <b>Traffic management for CAVs: Influence on traffic characteristics and control strategies</b><br>BEIJING UNIVERSITY OF TECHNOLOGY, CHINA<br>Invited seminar                                                                                                                               | 28/05/2019 |
| <b>User-based signal timing in a connected vehicle environment</b><br>ZHEJIANG UNIVERSITY, CHINA<br>Invited seminar                                                                                                                                                                         | 31/05/2019 |
| <b>Feedback-based integrated lane-based control with Connected and Automated Vehicles</b><br>UNIVERSITY OF CANTERBURY, NEW ZEALAND<br>Invited seminar                                                                                                                                       | 06/11/2019 |
| <b>Data-driven traffic flow estimation at intersections in a partially connected environment</b><br>IEEE INTELLIGENT TRANSPORTATION SYSTEM CONFERENCE 2020, GREECE<br>Workshop on Traffic Management for Future Mobility: CAVs in a Mixed Traffic Environment                               | 20/09/2020 |
| <b>Traffic Management in the era of Connectivity and Automation</b><br>TU DRESDEN, GERMANY<br>Invited lecture                                                                                                                                                                               | 08/10/2020 |
| <b>Online set-point estimation for motorway traffic control with connected and automated vehicles</b><br>UNIVERSITY OF CYPRUS, CYPRUS<br>Invited seminar                                                                                                                                    | 26/10/2022 |
| <b>Integration of large-scale activity-based and traffic assignment models: Calibration and equilibrium</b><br>QUEENSLAND UNIVERSITY OF TECHNOLOGY, AUSTRALIA<br>Invited seminar                                                                                                            | 18/10/2023 |
| <b>Connected and Automated Driving: Obstacle avoidance and cooperative decentralised routing</b><br>UNIVERSITY OF QUEENSLAND, AUSTRALIA<br>Invited seminar                                                                                                                                  | 23/10/2023 |
| <b>Connected and Automated Driving: Decentralised signal-free intersection management and cooperative routing</b><br>UNIVERSITY OF SYDNEY, AUSTRALIA<br>Invited seminar                                                                                                                     | 03/11/2023 |

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| <b>Dynamic ridesharing: Considering traffic congestion effects and fair pricing in matching and routing</b> | 13/11/2023 |
| UNIVERSITY OF NEW SOUTH WALES, AUSTRALIA                                                                    |            |
| Invited seminar                                                                                             |            |
| <b>Matching and routing in Dynamic Ridesharing: Addressing traffic congestion and fair pricing</b>          | 21/03/2024 |
| TU DELFT, THE NETHERLANDS                                                                                   |            |
| Invited webinar                                                                                             |            |
| <b>Controlling traffic when flow dynamics evolve</b>                                                        | 16/06/2026 |
| TRAFFIC AND GRANULAR FLOW 2026, UK                                                                          |            |
| Keynote speaker                                                                                             |            |

## Invited external examiner

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|--------------------------------------------------------------------|---------|
| <b>PhD degree examination</b>                                      | 12/2025 |
| UNIVERSITY OF ALCALÁ, SPAIN                                        |         |
| <b>PhD degree examination</b>                                      | 09/2025 |
| TECHNICAL UNIVERSITY OF CRETE, GREECE                              |         |
| <b>Faculty evaluator</b>                                           | 02/2025 |
| UNIVERSITÉ GUSTAVE EIFFEL, FRANCE                                  |         |
| <b>PhD degree examination</b>                                      | 09/2025 |
| TECHNICAL UNIVERSITY OF CRETE, GREECE                              |         |
| <b>PhD degree examination</b>                                      | 11/2024 |
| UNIVERSITY OF TARTU, ESTONIA                                       |         |
| <b>PhD degree examination</b>                                      | 05/2024 |
| TECHNION – ISRAEL INSTITUTE OF TECHNOLOGY, ISRAEL                  |         |
| <b>PhD degree examination</b>                                      | 05/2024 |
| TRINITY COLLEGE DUBLIN, IRELAND                                    |         |
| <b>PhD degree examination</b>                                      | 12/2022 |
| TECHNICAL UNIVERSITY OF CRETE, GREECE                              |         |
| <b>PhD entry examination</b>                                       | 12/2020 |
| SWISS FEDERAL INSTITUTE OF TECHNOLOGY IN ZURICH (ETH), SWITZERLAND |         |
| <b>PhD degree examination</b>                                      | 07/2019 |
| TECHNICAL UNIVERSITY OF CRETE, GREECE                              |         |
| <b>PhD degree examination</b>                                      | 09/2017 |
| TECHNICAL UNIVERSITY OF CRETE, GREECE                              |         |
| <b>PhD degree examination</b>                                      | 03/2017 |
| TECHNICAL UNIVERSITY OF CRETE, GREECE                              |         |
| <b>2 MSc degree theses</b>                                         | 02/2017 |
| ÉCOLE POLYTECHNIQUE FÉDÉRALE DE LAUSANNE (EPFL), SWITZERLAND       |         |

## Research evaluation

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- Evaluator of research proposals for California METRANS Transportation Center
- Evaluator of research proposals for the Netherlands Organisation for Scientific Research (NWO), domain Applied and Engineering Sciences
- Evaluator of research proposals for the Luxembourg national research fund (FNR)
- Evaluator of research proposals for the Research Grants Council (RGC) of Hong Kong
- Evaluator of research proposals for the Swiss National Science Foundation (SNSF)

- Evaluator of grant proposals for the National Science Center (NCN) of Poland
- External Expert in the evaluation of COST Action proposals, for the European Commission

## Recognitions and awards

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- The paper by Roncoli, C., Papageorgiou, M., Papamichail, M., titled “Traffic flow optimisation in presence of vehicle automation and communication systems - Part II: Optimal control for multi-lane motorways”, was listed among the “Most Cited Transportation Research Part C: Emerging Technologies Articles” during 2015-2018.
- The paper by Spiliopoulou, A., Perraki, G., Papageorgiou, M., Roncoli, C., titled “Exploitation of ACC systems towards improved traffic flow efficiency on motorways”, which was presented by A. Spiliopoulou at the 5th IEEE International Conference on Models and Technologies for Intelligent Transportation Systems (MT-ITS 2017), Naples, Italy, June 26-28, 2017, received the “Best Paper Presented by a Woman award”.
- The paper by Bekiaris-Liberis, N., Roncoli, C., Papageorgiou, M., titled “Highway Traffic State Estimation With Mixed Connected and Conventional Vehicles”, received the 2019 “George N. Saridis Best Transactions Paper Award” for Outstanding Research.
- The paper by Zhou, Z., Roncoli, C., titled “A fairness-aware joint pricing and matching framework for dynamic ridesharing”, which was presented by Z. Zhou at the 8th IEEE International Conference on Models and Technologies for Intelligent Transportation Systems (MT-ITS 2023), Nice, France, June 14-16, 2023, received the “Best Paper Award”.
- The paper by Vitale, F., Roncoli, C., titled “Real-time congestion-aware distributed cooperative rerouting of connected and automated vehicles in urban networks”, which was presented by F. Vitale at the TRB 103rd Annual Meeting, Washington, DC, USA, January 7-11, 2024, received the “Best Young First Author Award” for the Operations Section.

## Personal information

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- Nationality: Italian
- Date of birth: 30/10/1984
- Languages: Italian (mother tongue), English (fluently), Dutch (intermediate), French (basic), Greek (basic)

### ARTICLES IN INTERNATIONAL JOURNALS

1. Roncoli, C., Bersani, C., and Sacile, R. A risk-based system of systems approach to control the transport flows of dangerous goods by road. *IEEE Systems Journal*, 7(4):561–570, 2013. DOI: 10.1109/JSYST.2012.2212652.
2. Iordanidou, G.-R., Roncoli, C., Papamichail, I., and Papageorgiou, M. Feedback-based mainstream traffic flow control for multiple bottlenecks on motorways. *IEEE Transactions on Intelligent Transportation Systems*, 16(2):1–12, 2014. DOI: 10.1109/TITS.2014.2331985.
3. Pasquale, C., Papamichail, I., Roncoli, C., Sacone, S., Siri, S., and Papageorgiou, M. Two-class freeway traffic regulation to reduce congestion and emissions via nonlinear optimal control. *Transportation Research Part C: Emerging Technologies*, 55(1):85–99, 2015. DOI: 10.1016/j.trc.2015.01.013.
4. Roncoli, C., Papageorgiou, M., and Papamichail, I. Traffic flow optimisation in presence of vehicle automation and communication systems – Part I: A first-order multi-lane model for motorway traffic. *Transportation Research Part C: Emerging Technologies*, 57:241–259, 2015. DOI: 10.1016/j.trc.2015.06.014.
5. Roncoli, C., Papageorgiou, M., and Papamichail, I. Traffic flow optimisation in presence of vehicle automation and communication systems – Part II: Optimal control for multi-lane motorways. *Transportation Research Part C: Emerging Technologies*, 57:260–275, 2015. DOI: 10.1016/j.trc.2015.05.011.
6. Roncoli, C., Papamichail, I., and Papageorgiou, M. Hierarchical model predictive control for multi-lane motorways in presence of Vehicle Automation and Communication Systems. *Transportation Research Part C: Emerging Technologies*, 62:117–132, 2016. DOI: 10.1016/j.trc.2015.11.008.
7. Bekiaris-Liberis, N., Roncoli, C., and Papageorgiou, M. Highway traffic state estimation with mixed connected and conventional vehicles. *IEEE Transactions on Intelligent Transportation Systems*, 17(12):3484–3497, 2016. DOI: 10.1109/TITS.2016.2552639.
8. Roncoli, C., Bekiaris-Liberis, N., and Papageorgiou, M. Use of speed measurements for highway traffic state estimation: Case studies on NGSIM data and highway A20, Netherlands. *Transportation Research Record: Journal of the Transportation Research Board*, 2559(1):90–100, 2016. DOI: 10.3141/2559-11.
9. Han, Y., Hegyi, A., Yuan, Y., Hoogendoorn, S., Papageorgiou, M., and Roncoli, C. Resolving freeway jam waves by discrete first-order model-based predictive control of variable speed limits. *Transportation Research Part C: Emerging Technologies*, 77:405–420, 2017. DOI: 10.1016/j.trc.2017.02.009.
10. Fountoulakis, M., Bekiaris-Liberis, N., Roncoli, C., Papamichail, I., and Papageorgiou, M. Highway traffic state estimation with mixed connected and conventional vehicles: Microscopic simulation-based testing. *Transportation Research Part C: Emerging Technologies*, 78:13–33, 2017. DOI: 10.1016/j.trc.2017.02.015.
11. Roncoli, C., Bekiaris-liberis, N., and Papageorgiou, M. Lane-changing feedback control for efficient lane assignment at motorway bottlenecks. *Transportation Research Record: Journal of the Transportation Research Board*, 2625:20–31, 2017. DOI: 10.3141/2625-03.
12. Iordanidou, G.-R., Papamichail, I., Roncoli, C., and Papageorgiou, M. Feedback-based integrated motorway traffic flow control with delay balancing. *IEEE Transactions on Intelligent Transportation Systems*, 18(9): 2319–2329, 2017. DOI: 10.1109/TITS.2016.2636302.
13. Bekiaris-Liberis, N., Roncoli, C., and Papageorgiou, M. Highway traffic state estimation per lane in the presence of connected vehicles. *Transportation Research Part B: Methodological*, 106:1–28, 2017. DOI: 10.1016/j.trb.2017.11.001.
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15. Papadopoulou, S., Roncoli, C., Bekiaris-Liberis, N., Papamichail, I., and Papageorgiou, M. Microscopic simulation-based validation of a per-lane traffic state estimation scheme for highways with

- connected vehicles. *Transportation Research Part C: Emerging Technologies*, 86:441–452, 2018. DOI: 10.1016/j.trc.2017.11.012.
16. Perraki, G., Roncoli, C., Papamichail, I., and Papageorgiou, M. Evaluation of a model predictive control framework for motorway traffic involving conventional and automated vehicles. *Transportation Research Part C: Emerging Technologies*, 92:456–471, 2018. DOI: 10.1016/j.trc.2018.05.002.
  17. Roncoli, C., Bekiaris-Liberis, N., Carlson, R. C., and Geroliminis, N. Special issue on “Management of Future Motorway and Urban Traffic Systems”. *Transportation Research Part C: Emerging Technologies*, 96:1–2, 2018. DOI: 10.1016/j.trc.2018.09.005.
  18. Bekiaris-Liberis, N., Roncoli, C., and Papageorgiou, M. Predictor-Based Adaptive Cruise Control Design. *IEEE Transactions on Intelligent Transportation Systems*, 19(10):3181–3195, 2018. DOI: 10.1109/TITS.2017.2771501.
  19. Han, Y., Hegyi, A., Yuan, Y., Roncoli, C., and Hoogendoorn, S. An extended Linear Quadratic Model Predictive Control approach for multi-destination urban traffic networks. *IEEE Transactions on Intelligent Transportation Systems*, 20(10):3647–3660, 2019. DOI: 10.1109/TITS.2018.2877259.
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  21. Papamichail, I., Bekiaris-Liberis, N., Delis, A. I., Manolis, D., Mountakis, K.-S., Nikolos, I. K., Roncoli, C., and Papageorgiou, M. Motorway traffic flow modelling, estimation and control with vehicle automation and communication systems. *Annual Reviews in Control*, 48:325–346, 2019. DOI: 10.1016/j.arcontrol.2019.09.002.
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  23. Bagheri, M., Mladenovic, M. N., Kosonen, I., Nurminen, J. K., Roncoli, C., and Ylä-Jääski, A. A computational framework for revealing competitive travel times with low-carbon modes based on smartphone data collection. *Journal of Advanced Transportation*, 2020:1–20, 2020. ISSN 0197-6729. DOI: 10.1155/2020/4693750.
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  27. Tajdari, F., Roncoli, C., and Papageorgiou, M. Feedback-based ramp metering and lane-changing control with connected and automated vehicles. *IEEE Transactions on Intelligent Transportation Systems*, 23(2), 2020. DOI: 10.1109/TITS.2020.3018873.
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  29. Zhao, M., Roncoli, C., Wang, Y., Bekiaris-Liberis, N., Guo, J., and Cheng, S. Generic approaches to estimating freeway traffic state and percentage of connected vehicles with fixed and mobile sensing. *IEEE Transactions on Intelligent Transportation Systems*, 23(8), 2021. DOI: 10.1109/TITS.2021.3121181.
  30. Mohammadi, R. and Roncoli, C. Towards data-driven vehicle estimation for signalised intersections in a partially connected environment. *Sensors*, 21(24), 2021. DOI: 10.3390/s21248477.

31. Han, Y., Wang, M., Li, L., Gao, J., Roncoli, C., and Liu, P. A physics-informed reinforcement learning-based strategy for local and coordinated ramp metering. *Transportation Research Part C: Emerging Technologies*, 137:103584, 2022. DOI: 10.1016/j.trc.2022.103584.
32. Agriesti, S. A. M., Roncoli, C., and Nahmias-Biran, B.-H. Assignment of synthetic population for activity-based modelling employing publicly available data. *ISPRS International Journal of Geo-Information*, 11(2):148, 2022. DOI: 10.3390/ijgi11020148.
33. Zhou, Z. and Roncoli, C. A scalable vehicle assignment and routing strategy for real-time on-demand ridesharing considering endogenous congestion. *Transportation Research Part C: Emerging Technologies*, 139:103658, 2022. DOI: 10.1016/j.trc.2022.103658.
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37. Tajdari, F. and Roncoli, C. Online set-point estimation for feedback-based traffic control applications. *IEEE Transactions on Intelligent Transportation Systems*, 24(10):10830–10842, 2023. DOI: 10.1109/TITS.2023.3274233.
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40. Vitale, F. and Roncoli, C. Reference tracking optimization with obstacle avoidance via task prioritization for automated driving. *IEEE Transactions on Intelligent Transportation Systems*, 25(2):1200–1214, 2023. DOI: 10.1109/TITS.2023.3312999.
41. Zhou, Z., Roncoli, C., and Sipetas, C. Optimal matching for coexisting ride-hailing and ridesharing services considering pricing fairness and user choices. *Transportation Research Part C: Emerging Technologies*, 156: 104326, 2023. DOI: 10.1016/j.trc.2023.104326.
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53. Vitale, F. and Roncoli, C. Cooperative traffic dispersion through rerouting of connected and automated vehicles in urban networks. *Transportation Research Record: Journal of the Transportation Research Board*, 2678 (12):980–998, 2024. DOI: 10.1177/03611981241248648.
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55. Agriesti, S. A. M., Roncoli, C., and Nahmias-Biran, B.-h. A simulation-based framework for quantifying potential demand loss due to operational constraints in automated mobility services. *Transportation Research Part A: Policy and Practice*, 192:104372, 2025. DOI: 10.1016/j.tra.2024.104372.
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69. Niroumand, R., Vosough, S., Roncoli, C., Rinaldi, M., and Connors, R. Mitigating network traffic congestion via link- and path-based incentives. Submitted.
70. Niroumand, R., Kafashan, F., Hajibabai, L., Hajbabaie, A., and Roncoli, C. A multi-trajectory construction heuristic to manage CAVs in signal-free intersections. Submitted.
71. Lee, E., Agriesti, S., Vosough, S., Sipetas, C., Roncoli, C., Rinaldi, M., and Leclercq, L. Unpacking seamlessness in transportation: A systematic literature review approach. Submitted.
72. Huang, X., Yao, S., Zhou, Q., Kaparias, I., Roncoli, C., and Hu, S. Predicting traffic flow using a dynamic temporal multi-graph convolution network with learnable propagation operators. Submitted.
73. Haris, M. and Roncoli, C. Efficient MILP-based urban network traffic control in mixed autonomy with dynamic saturation rates. Submitted.
74. Wang, H. and Roncoli, C. Network-wide fine-grained passenger loading prediction via dynamic relational spatiotemporal deep graph neural networks. Submitted.
75. Lyu, H., Roncoli, C., Liu, Z., and Guo, Y. A prediction-guided physics-encoded deep learning framework for car-following simulation in mixed-autonomy traffic. Submitted.
76. Yang, Y., Roncoli, C., Zhu, J., Qian, Y., and Yang, X. Physics-informed meta-learning for traffic state prediction in urban road networks: Encapsulating traffic flow theory into neural networks. Submitted.

#### ARTICLES IN CONFERENCE PROCEEDINGS

1. Roncoli, C., Bersani, C., Sacile, R., and Minciardi, R. Optimal traffic flow assignment for planned deliveries in dangerous good transportation. In *2011 IEEE International Systems Conference*, 292–297, 2011. DOI: 10.1109/SYSCON.2011.5929051.
2. Roncoli, C., Bersani, C., and Sacile, R. A decentralised control model for the risk-based management of dangerous good transport flows on road. In *2012 IEEE International Systems Conference SysCon 2012*, 1–5, 2012. DOI: 10.1109/SysCon.2012.6189522.
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7. Laarabi, M. H., Roncoli, C., Sacile, R., Boulmakoul, A., and Garbolino, E. An overview of a multiagent-based simulation system for dynamic management of risk related to Dangerous Goods Transport. In *2013 IEEE International Systems Conference (SysCon)*, 830–835, 2013. DOI: 10.1109/SysCon.2013.6549980.
8. Roncoli, C., Papageorgiou, M., and Papamichail, I. Optimal control for multi-lane motorways in presence of vehicle automation and communication systems. *IFAC Proceedings Volumes*, 47(3):4178–4183, 2014. DOI: 10.3182/20140824-6-ZA-1003.01657.
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11. Roncoli, C., Papamichail, I., and Papageorgiou, M. Model predictive control for motorway traffic with mixed manual and VACS-equipped vehicles. *Transportation Research Procedia*, 10:452–461, 2015. DOI: 10.1016/j.trpro.2015.09.095.
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